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Can Intel challenge TSMC's advanced packaging leadership?

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Intel's EMIB-T vs TSMC's CoPoS—the race is on

Advanced packaging is key to scaling Cloud AI performance while improving power efficiency. The advanced packaging TAM is growing rapidly and could reach US\$122bn by 2030E, from US\$19bn in 2026E, a 59% CAGR. TSMC's CoWoS (chip on wafer on substrate) is the current mainstream for Cloud AI accelerators, but as packages increase in size and complexity, the industry needs new solutions. Intel's EMIB-T is emerging as a competitive alternative given TSMC's supply constraints and growing interest in supply-chain diversification, while TSMC is advancing CoPoS (chip on panel on substrate). This report examines competitive strengths and supply chain opportunities for both.

EMIB-T ideal for scalability & cost; CoPoS offers better performance

EMIB-T and CoPoS are both credible next-gen advanced packaging platforms for large AI accelerators, but they are optimized for different priorities. **EMIB-T's** core value proposition is its simpler bridge-based architecture, competitive structural cost through elimination of the interposer, and greater scalability to ultra-large package sizes by building directly on large substrates. By contrast, TSMC's **CoPoS** should offer stronger package-level performance through tighter interconnect, better signal integrity, broader integration options such as 3D stacking (and eventually, CPO), and TSMC's execution track record. Key challenges: Intel must lift EMIB-T packaging from ~90–92% validation yields toward high-volume manufacturing levels, while TSMC must overcome CoPoS warpage, panel process uniformity, and new equipment readiness. EMIB-T "substrate" fabrication yields may currently be at ~50%, vs >80% for HPC ABF substrates, but capacity expansion at Ibiden could offset lower yields and support selected products like Google's TPU v9, in our view. Further challenges for EMIB-T are integration between the foundry (TSMC) and back-end (Intel) processes, as well as yield responsibility.

TSMC remains well positioned, with robust growth ahead

We believe EMIB-T could be a meaningful second-source for selected hyperscaler ASICs, although CoPoS should remain the preferred platform for customers prioritizing the highest performance and execution reliability. EMIB-T aims to enter mass production in H227-28, slightly ahead of TSMC's CoPoS in 2028. We anticipate TSMC to maintain high market share, assuming CoPoS enters mass production for Nvidia in 2028, and for others in 2029-30. In our stress test, assuming Intel gains a meaningful 13% industry share or US\$16bn sales by 2030, equivalent to packages for ~24mn accelerator dies, we estimate TSMC's advanced packaging sales would deliver a 50% CAGR from 2026-30, with 65-70% market share by 2030E. Advanced packaging would account for 18% of TSMC sales in 2030E, up from 9% in 2026E. In our lower case, which assumes stronger adoption of Intel's EMIB-T, we estimate TSMC's advanced packaging sales at a slower 47% CAGR in 2026-30E, reaching 16% of sales in 2030E with 62% market share.

Stock implications

We think TSMC (Buy) remains well positioned for the Cloud AI opportunity, and should retain its tech leadership in advanced packaging. While a successful EMIB-T ramp could lift Intel's foundry sales and profitability, we remain Neutral due to valuation considerations. We like MediaTek on its TPU opportunity and see the EMIB-T execution risks as manageable. CoPoS and EMIB-T should drive another capex cycle, benefiting equipment suppliers. We initiate [All Ring](#) at Buy with a PT of NT\$1,550. We also like GPTC, Chroma, ASMPT, and Gudeng. Substrate makers should also benefit from content increase; our top picks are Ibiden and Unimicron. See inside for [Major supply chain beneficiaries](#).

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Figure 1: Valuation summary of major companies related to CoPoS and EMIB-T

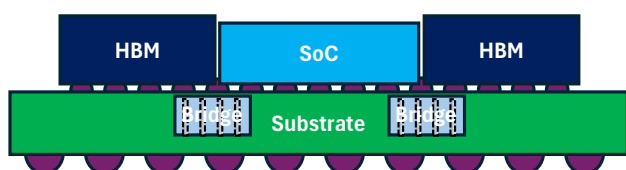
Company name	Ticker	Rating	Analyst	Market cap (US\$m)	Price target (LC)	Share price (LC)	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)		Dividend yield (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Foundry																
TSMC	2330.TW	Buy	Sunny Lin	1,931,966	3,650	2,350	64.5	36.9	21.6	15.8	8.0	5.8	43.4	42.7	0.9	1.3
Intel	INTC.O	Neutral	Timothy Arcuri	521,617	121	100	233.8	37.7	70.4	51.1	4.3	3.6	6.2	7.6	0.0	0.0
Average									46.0	33.4	6.1	4.7				
Fabless																
MediaTek	2454.TW	Buy	Sunny Lin	192,527	6,500	3,750	(3.4)	183.2	58.7	20.7	15.2	11.3	25.7	62.7	1.4	1.4
Broadcom	AVGO.O	Buy	Timothy Arcuri	1,881,276	485	392	67.3	80.9	32.8	18.1	17.4	10.8	60.6	74.0	0.6	0.7
Average									45.7	19.4	16.3	11.0				
Semi-backend & equipment																
All Ring	6187.TWO	Buy	Sunny Lin	3,091	1,550	942	81.7	60.8	33.6	20.9	11.2	8.6	34.9	46.2	1.1	1.9
Amkor	AMKR.O	Neutral	Randy Abrams	16,449	80	65	45.7	25.3	29.8	23.8	3.3	2.9	11.6	13.1	0.7	0.9
ASE	3711.TW	Buy	Sunny Lin	89,422	835	613	87.6	66.5	34.9	21.0	6.9	5.8	20.9	30.0	1.0	1.9
ASMPT	0522.HK	Buy	Sunny Lin	8,802	200	167	106.1	41.2	31.7	22.4	3.8	3.5	12.4	16.1	0.8	1.6
Chroma ATE	2360.TW	Buy	Jerry Su	28,401	2,960	2,080	63.5	37.6	45.9	33.4	20.5	15.8	51.2	53.4	0.4	0.9
Gudeng	3680.TWO	Buy	Sunny Lin	1,463	710	460	26.5	67.6	38.6	23.0	3.8	3.5	10.7	15.8	1.5	1.9
GPTC	3131.TWO	Buy	Sunny Lin	2,711	5,000	2,865	72.6	57.5	36.5	23.2	13.6	10.2	41.7	50.3	1.1	1.9
Powertech	6239.TW	Neutral	Jerry Su	6,914	240	280	66.5	20.8	22.5	18.6	3.3	3.0	15.5	16.9	3.0	1.6
Average									34.3	23.6	7.9	6.4				
Substrates																
Ibiden	4062.T	Buy	Shingo Hirata, CFA	30,949	20,500	18,075	72.4	48.8	64.3	43.2	8.2	7.0	8.8	13.4	0.2	0.2
Unimicron	3037.TW	Buy	Diana Chang	42,294	1,200	839	260.4	98.8	60.6	30.5	10.7	8.4	19.2	30.8	0.2	0.2
Average									62.4	36.8	9.4	7.7				

Source: UBS estimates. Note: Taiwan stocks priced as of 24 July 2026; non-Taiwan stocks priced as of 23 July 2026.

Key insights & conclusions

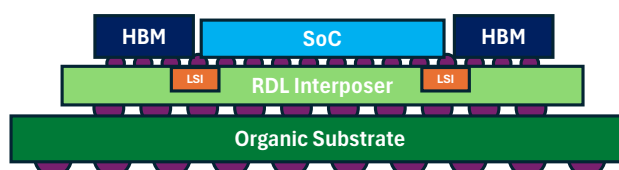
First, rather than representing a thesis-changing threat to TSMC, we believe the emergence of Intel's EMIB-T should be viewed as validation of the structural growth opportunity in advanced packaging. As AI accelerators continue to scale in die count, HBM content and power consumption, leading-edge logic alone is no longer sufficient to drive system-level performance gains. Both EMIB-T and CoPoS aim to address the industry's next bottleneck—package size, interconnect density and power delivery—but they do so through different architectural approaches. Intel is pursuing a more substrate-centric and cost-efficient route that prioritizes scalability, while TSMC is extending its interposer-based ecosystem to maximize integration density and package-level performance. We expect the two technologies to coexist over time, serving different customer requirements and optimization priorities.

Figure 2: EMIB-T structure more simplified, without interposer



Source: UBS

Figure 3: CoPoS structure



Source: UBS

Second, our industry work suggests CoPoS retains structural advantages in performance and ecosystem maturity, while EMIB-T's key differentiation lies in cost efficiency and scalability. TSMC's interposer-based architecture should continue to enable superior interconnect density, signal integrity, power integrity and future integration opportunities, including embedded capacitors, advanced 3D stacking and co-packaged optics. By contrast, EMIB-T's elimination of the interposer and its direct integration of silicon bridges into the substrate could provide a more economical pathway toward ultra-large AI packages, while alleviating some of the scaling challenges associated with very large interposer structures, such as package warpage. We think the key question for both technologies has shifted from theoretical capability to manufacturing execution. Intel must demonstrate sustained high-volume yields for EMIB-T, while TSMC must successfully industrialize panel-level processing and manage warpage at increasingly larger package dimensions.

Figure 4: Intel EMIB-T vs. TSMC CoPoS - strengths and weaknesses

Category	Intel EMIB-T	TSMC CoPoS	Who wins
Architecture	Direct die attach on substrate with localized silicon bridges	Die attach on large interposer / panel before substrate integration	EMIB-T simpler structure
Performance	Strong chiplet connectivity, improved power delivery via TSV bridges	Denser interconnect, better signal integrity, richer package integration	CoPoS
Power Integrity	TSV-enabled vertical power delivery; embedded MIM capacitors	More room for embedded capacitors (eDTC / iCAP) and power stabilization	CoPoS
Package Complexity	Fewer process steps, no full-size interposer	More complex package stack and manufacturing flow	EMIB-T
Scalability (>12x Reticle)	Well suited for ultra-large packages using substrate scaling	Scaling constrained by interposer/panel warpage challenges	EMIB-T
Manufacturing Risk	Yield ramp for TSV bridges and large substrates remains key challenge	Warpage and new panel-level tool development are major hurdles	Tie
Yield Confidence	Validation yield ~90%+, but mass-production yield still needs proving	TSMC has strongest industry track record in packaging execution	CoPoS
Cost Potential	Lower BOM due to no large interposer and simpler flow	Higher structural cost but may improve with panel economics	EMIB-T assuming reasonable yield
Time-to-Market	Targets H227 commercial ramp	Targets 2028 mass production	EMIB-T
Ecosystem Strength	Growing support from hyperscalers seeking alternative supply	Deep ecosystem spanning foundry, CoWoS, SoIC and future CPO	CoPoS
Best For	Cost-efficient scaling to larger AI packages	Maximum performance and highest integration density	Depends on customer priorities

Source: UBS

Finally, we remain constructive on TSMC's long-term advanced packaging outlook despite the emergence of credible alternatives. Even under a scenario where EMIB-T gains meaningful adoption among select hyperscaler ASIC programs, we expect TSMC to maintain its leadership position through its unique combination of leading-edge foundry manufacturing, broad advanced packaging portfolio and track record of successful volume ramp execution. We view CoPoS as a full-generation upgrade to CoWoS rather than merely a packaging form-factor transition, positioning TSMC to continue supporting the industry's roadmap toward larger, more power-intensive AI systems. In our view, the most likely outcome is not a winner-takes-all market, but rather

a more diversified advanced packaging landscape in which Intel establishes itself as a meaningful secondary source while TSMC retains the dominant share of the highest-performance AI packaging opportunity through the end of the decade.

Figure 5: Stress test scenario—TSMC's advanced packaging could sustain 65-70% share by 2030E, even if Intel ramps EMIB-T to support 24mn XPU dies or 6mn chipsets

(US\$mn)	2026E	2027E	2028E	2029E	2030E	26-30E CAGR
Advanced packaging industry revenue	19,025	36,300	58,080	87,120	121,968	59%
YoY	87%	91%	60%	50%	40%	
TSMC's advanced packaging sales	16,034	29,730	41,676	59,207	81,378	50%
YoY	80%	85%	40%	42%	37%	
% of TSMC total sales	9%	12%	13%	16%	18%	
Intel's EMIB-T sales for external			5,438	9,570	15,791	
Google TPU			5,003	8,374	13,159	
Units (mn)			2.3	3.5	5.0	
EMIB-T content (US\$)			2,175	2,393	2,632	
Others			435	1,196	2,632	
Units (mn)			0.2	0.5	1.0	
EMIB-T content (US\$)			2,175	2,393	2,632	
Advanced packaging industry share						
TSMC	84%	82%	72%	68%	67%	
Intel			9%	11%	13%	
OSATs	16%	18%	19%	21%	20%	

Source: UBS estimates

Figure 6: Lower case scenario - TSMC's share in advanced packaging could decline to 62% by 2030E, assuming better success of Intel's EMIB-T

(US\$mn)	2026E	2027E	2028E	2029E	2030E	26-30E CAGR
Advanced packaging industry revenue	19,025	36,300	58,080	86,883	120,081	59%
YoY	87%	91%	60%	50%	38%	
TSMC's advanced packaging sales	16,034	29,730	41,676	56,578	74,227	47%
YoY	80%	85%	40%	36%	31%	
% of TSMC total sales	9%	12%	13%	15%	16%	
Intel's EMIB-T sales for external			5,438	11,963	21,054	
Google TPU			5,003	10,766	17,106	
Units (mn)			2.3	4.5	6.5	
EMIB-T content (US\$)			2,175	2,393	2,632	
Others			435	1,196	3,948	
Units (mn)			0.2	0.5	1.5	
EMIB-T content (US\$)			2,175	2,393	2,632	
Advanced packaging industry share						
TSMC	84%	82%	72%	65%	62%	
Intel			9%	14%	18%	
OSATs	16%	18%	19%	21%	21%	

Source: UBS estimates

EMIB vs CoPoS

With the fast growing demand for Cloud AI compute, performance gains from advancement of front-end leading-edge process nodes alone are no longer sufficient. Advanced packaging has become an essential enabler of total chipset performance, facilitating high-speed interconnect between compute chips and memory. TSMC's CoWoS (chip-on-wafer-on-substrate) has been the industry's leading solution since the AI-driven surge in demand began in 2023. The technology has demonstrated proven yields and large volume production capability, and is widely adopted by major AI chip developers, including Nvidia, AMD, and Broadcom. However, due to persistent CoWoS capacity constraints at TSMC in recent years, other advanced packaging solutions from competing foundries and OSATs are gaining traction. Intel's EMIB (Embedded Multi-die Interconnect Bridge) technology has emerged as one of the most closely watched solutions for ultra-large packages that could become a meaningful competitor to TSMC

in 2028 and beyond. Meanwhile, TSMC has accelerated development of its CoPoS (chip on panel on substrate) platform, targeting mass production by 2028.

This report examines: **1)** the underlying technologies behind Intel's EMIB and TSMC's next-generation CoPoS; **2)** a comparison of the two approaches, including production challenges, scalability and cost competitiveness; **3)** the implications for TSMC's long-term advanced packaging growth outlook; **4)** the potential role of glass core substrates; and **5)** major beneficiaries across the supply chain.

What is EMIB?

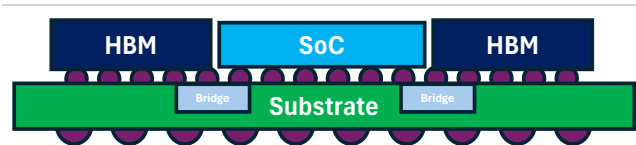
Intel’s **EMIB** (Embedded Multi-die Interconnect Bridge) is a 2.5D packaging technology that connects multiple chips/chiplets together within a single package, utilizing small silicon bridges embedded directly in the package substrate. Intel has been developing EMIB for more than a decade, but primarily for its own internal CPU products. Now, given the company's growing foundry ambitions and the rapid expansion of Cloud AI demand, Intel is accelerating the technology's development and investing to support external clients. A key advantage of EMIB compared to TSMC's CoWoS is that the compute dies / HBMs are attached directly to the substrate, eliminating the need for costly silicon interposers. This simpler architecture has the potential to lower package costs while reducing risks associated with coefficient of thermal expansion (CTE) properties and warpage.

The key determinant for EMIB yields and production scalability lies in manufacturing the critical substrate, which incorporates embedded silicon bridges. Intel defines the technology specs and reference process flow, then outsources the substrate manufacturing to packaging substrate makers. Ibiden in Japan has been Intel's longstanding partner on EMIB substrate, and in February 2026 announced a ¥220.0bn capex plan to expand its EMIB capacity, particularly for EMIB-T, targeting volume production in FY28. Historically, Intel's EMIB solution experienced low yields during early years of mass production, largely due to the complexity of substrate manufacturing.

One of the most challenging steps involved creating the high-precision cavities for silicon-bridge placement and integration which requires precise alignment bonding the silicon bridges. Our industry survey suggest that Intel and its substrate partners have accumulated extensive experience in cavity-related processes; however, this remains a key manufacturing challenge as substrate sizes increase substantially (potentially >9-10x reticle size for AI accelerators vs 2-3x today along with 20+ layer count). The difficulty is further amplified in EMIB-T, which requires a greater number of silicon bridges.

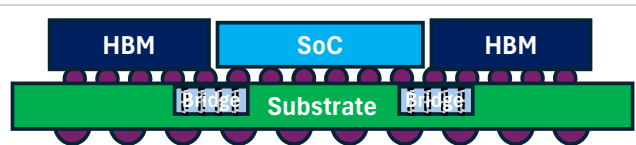
EMIB-T is Intel’s next-generation packaging technology, introduced in late 2024. The key enhancement is the incorporation of TSVs (through-silicon vias) directly through the embedded silicon bridges, enabling vertical power delivery from the bottom of the package to both the HBM stacks and the compute dies. This reduces the complexity of power routing, enabling EMIB-T to support the higher power requirements of next-generation AI chips and memory technologies, including HBM4 and more advanced memory generations in the future. Normal EMIB substrate yield should've reached more mature level at close to 60% level, but EMIB-T is still maturing so we estimate yields to be around 50% for the leading Japan suppliers.

Figure 7: EMIB with silicon bridge



Source: UBS

Figure 8: EMIB-T with silicon bridge and TSV for vertical power delivery



Source: UBS

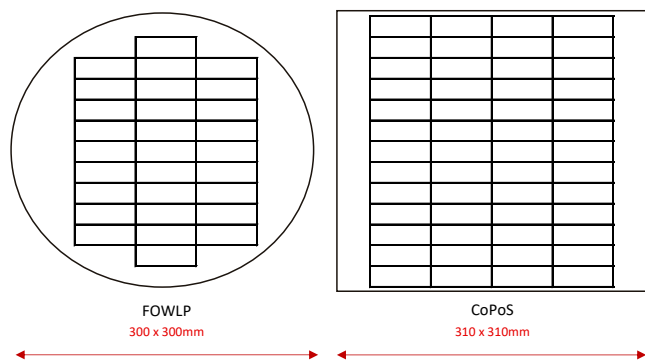
What is CoPoS?

CoPoS (chip on panel on substrate) is TSMC’s next-generation 2.5D advanced packaging technology, designed for ultra-large package sizes and positioned as the

successor to CoWoS (chip on wafer on substrate). One main goal of CoPoS is to transition from CoWoS's round-wafer format to a square/rectangular for the die attach on interposer, as CoWoS suffers from meaningful side-area loss when package size increases. Greater production efficiency and increased output would also be beneficial for yield management. Beyond enabling larger package sizes, our industry studies suggests that TSMC is developing CoPoS as a broader technology upgrade that could support higher interconnect density, more robust performance, and enhanced power integrity. In other words, CoPoS should be considered as a comprehensive upgrade of CoWoS across multiple dimensions, not just the form factor.

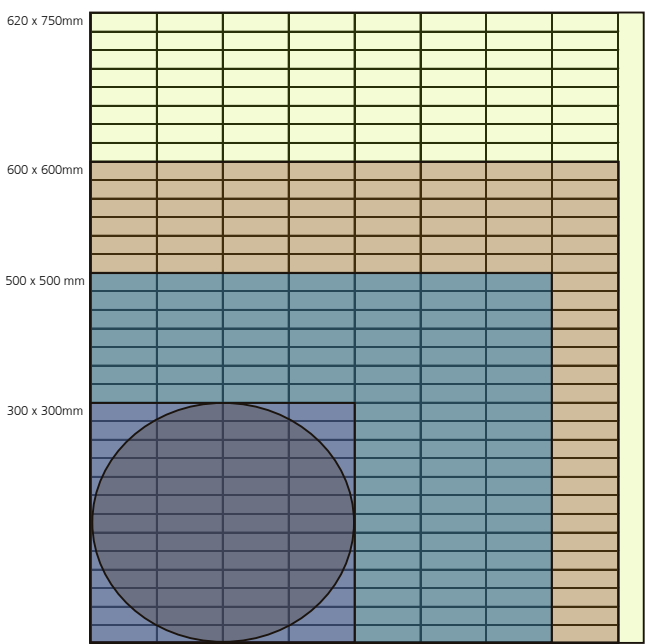
TSMC will likely start CoPoS ramp with 310x310mm² panels for the first generation, with mass production targeted for 2028. We believe Nvidia's Feynman may be the first CoPoS client in H228-2029, followed by AMD and ASICs in 2029-30. As the technology matures, we anticipate TSMC to migrate to larger panel sizes, such as 510x515mm², further enhancing scalability and production efficiency.

Figure 9: CoPoS (or panel-level packaging) has better area utilisation and production efficiency than CoWoS



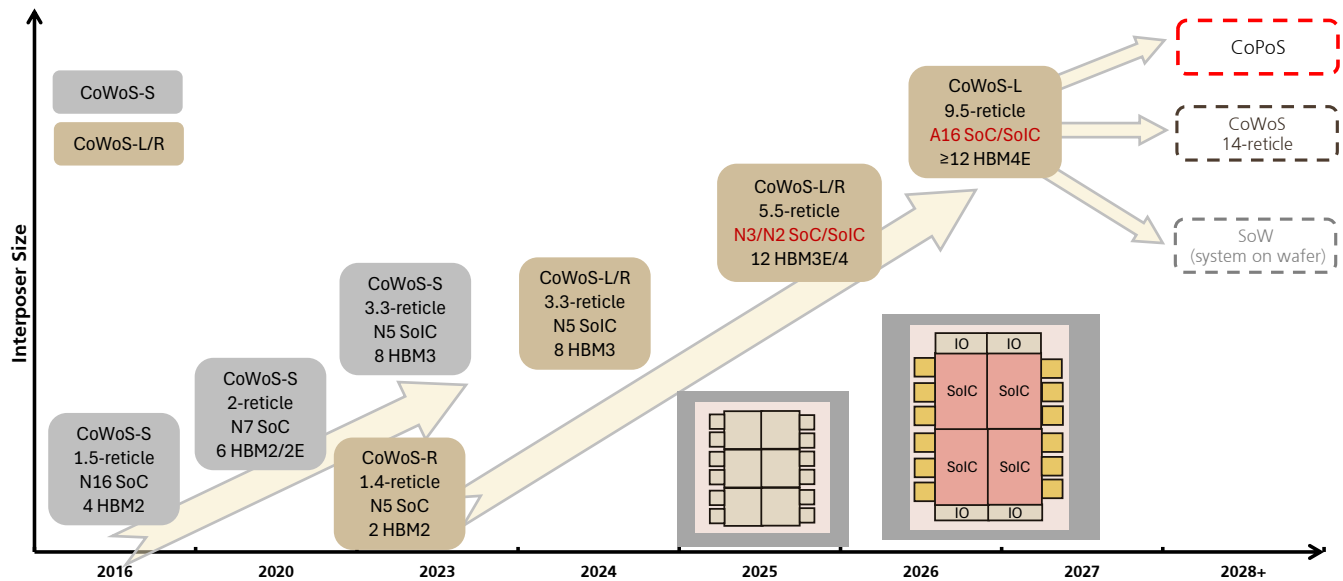
Source: UBS

Figure 10: CoPoS can achieve much larger interposer areas vs CoWoS



Source: UBS

Figure 11: TSMC's tech roadmap for advanced packaging



Source: Company data, UBS

Competitive analysis—EMIB-T vs CoPoS

Both Intel's EMIB-T and TSMC's CoPoS aim to be the key packaging solutions for the next generation of HPC chips. Below, we compare the two technologies across five key dimensions, highlighting their respective strengths, challenges and cost advantages, based on feedback from industry experts and equipment suppliers gathered to date.

Summary view: Intel's EMIB-T and TSMC's CoPoS are both viable next-generation advanced packaging platforms for large AI/HPC accelerators, but they are optimized for different priorities. EMIB-T's core value proposition is a simpler bridge-based architecture, lower structural cost due to the absence of a full silicon interposer, and potentially easier scaling to ultra-large package sizes by building directly on large substrates. TSMC's CoPoS, by contrast, should offer stronger package-level performance through higher interconnect density, better signal/power integrity, richer integration options such as embedded capacitors, 3D stacking and eventually co-packaged optics (CPO)—all with the support of TSMC's well-established manufacturing ecosystem. The key execution challenges differ: Intel must demonstrate that EMIB-T can move from ~90–92% validation yields toward high-volume manufacturing levels, while TSMC must overcome challenges related to CoPoS warpage, panel process uniformity, and new equipment/tool-chain readiness. Overall, we believe EMIB-T can become a meaningful second-source and cost/scaling alternative for selected hyperscaler ASICs, but that CoPoS is likely to remain the preferred platform for customers prioritizing the highest levels of performance and execution certainty.

Figure 12: Intel EMIB-T vs. TSMC CoPoS - strengths and weaknesses

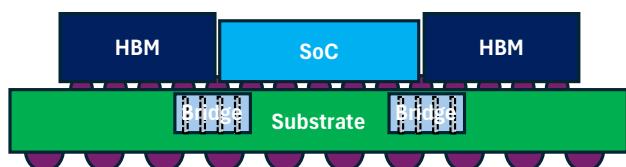
Category	Intel EMIB-T	TSMC CoPoS	Who wins
Architecture	Direct die attach on substrate with localized silicon bridges	Die attach on large interposer / panel before substrate integration	EMIB-T simpler structure
Performance	Strong chiplet connectivity, improved power delivery via TSV bridges	Denser interconnect, better signal integrity, richer package integration	CoPoS
Power Integrity	TSV-enabled vertical power delivery; embedded MIM capacitors	More room for embedded capacitors (eDTC / iCAP) and power stabilization	CoPoS
Package Complexity	Fewer process steps, no full-size interposer	More complex package stack and manufacturing flow	EMIB-T
Scalability (>12x Reticle)	Well suited for ultra-large packages using substrate scaling	Scaling constrained by interposer/panel warpage challenges	EMIB-T
Manufacturing Risk	Yield ramp for TSV bridges and large substrates remains key challenge	Warpage and new panel-level tool development are major hurdles	Tie
Yield Confidence	Validation yield ~90%+, but mass-production yield still needs proving	TSMC has strongest industry track record in packaging execution	CoPoS
Cost Potential	Lower BOM due to no large interposer and simpler flow	Higher structural cost but may improve with panel economics	EMIB-T assuming reasonable yield
Time-to-Market	Targets H227 commercial ramp	Targets 2028 mass production	EMIB-T
Ecosystem Strength	Growing support from hyperscalers seeking alternative supply	Deep ecosystem spanning foundry, CoWoS, SolC and future CPO	CoPoS
Best For	Cost-efficient scaling to larger AI packages	Maximum performance and highest integration density	Depends on customer priorities

Source: UBS

1) Technology structure

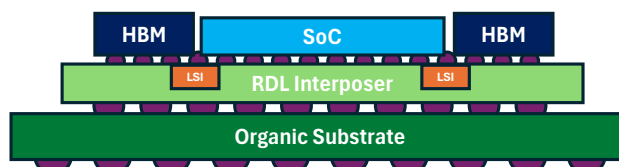
The key difference between Intel's EMIB-T and TSMC's CoPoS lies in the underlying packaging architecture. EMIB-T attaches compute dies, HBM and other components directly onto the substrate, which is more similar to traditional packaging. In contrast, CoPoS, which is more similar to CoWoS, places the die on an "interposer", which is then attached to the substrate. As a result, EMIB-T features a simpler packaging design and manufacturing process than CoWoS or CoPoS.

Figure 13: EMIB-T structure more simplified, without interposer



Source: UBS

Figure 14: CoPoS structure

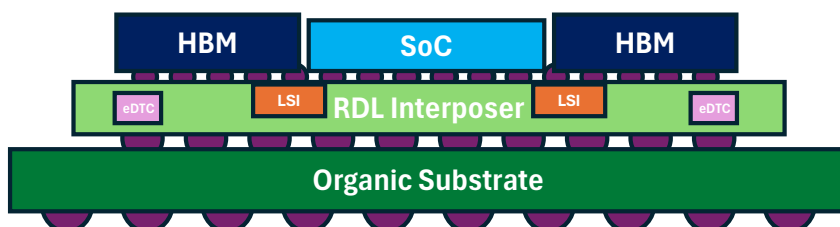


Source: UBS

2) Performance

Contrary to the view that TSMC's CoPoS is merely a form-factor change from CoWoS (ie, from circular to square carrier), we think CoPoS represents a compelling advanced packaging upgrade when it ramps in 2028. As CoPoS will still utilize a fine-pitch interposer within the package, it should offer more robust performance compared to EMIB-T, including support more complicated package structures, higher interconnect density between compute dies, memory and other components, as well as improved signal integrity. Meanwhile, TSMC could also incorporate a higher number of embedded deep-trench capacitors (eDTC)/integrated capacitors (iCAP) within/underneath the interposer, helping to suppress electrical noise and stabilise power within the package.

Figure 15: Illustration of embedded DTC inside/underneath the interposer



Source: UBS

Compared with standard EMIB, Intel's EMIB-T addresses HBM power-delivery limitations by incorporating TSVs through the silicon bridges. However, in our view, it is still likely to trail TSMC's CoWoS and CoPoS in terms of package performance, interconnect density, etc.

3) Key manufacturing challenges

Both technologies remain in the early stages of development and have yet to reach full manufacturing maturity. Intel EMIB-T's development appears to be slightly ahead in development and is slated for mass production in late 2027, with Google's TPU v9 expected to be its first major external product. We believe TSMC is targeting CoPoS mass production in 2028 and began by setting up a pilot line for technology evaluation and qualification in June 2026.

For **Intel's EMIB-T**, our conversations with industry experts suggest that although standard EMIB yield is high (ie >90%), EMIB-T may require additional time to reach

comparable levels. The experts believe that EMIB-T yields may not be fully optimised yet even with the initial production ramp slated to begin in late 2027-28. Our understanding is EMIB-T substrate yields are currently around 50%, while packaging yields are about 90%, based on trial production runs of MediaTek's TPU v9 prototype. For HPC products, however, packaging yields ideally should improve to the high 90% range, considering the substantial value of the compute chips and HBMs being integrated.

Key challenges for **EMIB-T** include:

1. EMIB-T is designed for much larger substrate sizes, ie 5x or even larger vs current EMIB package. The increased substrate area, combined with a much higher number of silicon bridges, raises the production difficulty.

2. More complex silicon bridge integration. EMIB-T's key upgrade is it builds TSV within the silicon bridge to enable vertical power delivery. This requires the formation of precise metal contacts at both the top and bottom of the substrate cavities, as well as highly accurate placement of the silicon bridges to ensure reliable power connections. Substrate makers may face a steeper learning curve than leading foundries like TSMC, making close collaboration with Intel and partners such as Ibiden critical to successful ramp-up.

3. According to industry experts, TC NCF (thermal compression with non-conductive film) process will be required for bridge attachment on the panel-level substrate cavities, rather than a standard die-attach process. This will require equipment suppliers to optimise the TC NCF bonders based on large-format panel production (510x515mm²) for substrate.

For **CoPoS**, major challenges include:

1. Warpage remains the most critical technical bottleneck as CoPoS retains an interposer-based architecture. As package size scales to over 9-10x reticles, warpage is expected to become increasingly severe due to the increase of surface area, in addition to the different coefficient of thermal expansion (CTE) of various materials (e.g. glass carrier and RDL interposer; substrate; PCB) used in the chip stacking process.

2. Equipment suppliers must develop an entirely new toolset to support the transition from CoWoS's circular-wafer format to CoPoS's square-panel format. The first-generation technology is expected to use 310x310mm² panels, increasing to 510x515mm² or larger as the technology matures. Most equipment will need to be re-designed to accommodate square-interposer manufacturing and chip-on-panel packaging. For instance, applying chemicals evenly during wet processes is more challenging on a square panel than on a circular wafer. One potential solution is to add more spray heads to ensure consistent coverage and uniformity across the entire panel surface.

4) Scalability for ultra-large package sizes

On scaling to ultra-large package sizes (eg, 12-14x or larger), feedback from experts suggests that Intel's EMIB-T could hold an edge over TSMC's CoPoS. This is mainly due to CoPoS's more complex packaging structure and the use of an interposer, which is generally more difficult to scale than a substrate-based technology. Intel's embedded interconnect bridge IP is a key enabler for Intel to build large chip packages directly on panel-sized substrates, without requiring ultra-large interposers. This approach could offer advantages in warpage control and thermal management. In addition, Intel can leverage its SMT (surface-mount technology) expertise to support chipmakers and hyperscalers with board-level integration of the packaged chips to help address warpage issues.

As highlighted in our [TSMC 2026 technology symposium takeaways](#), TSMC has added 14x-reticle CoWoS to its roadmap, with support for around 10 large compute dies and 20 HBM stacks. The technology is slated for production in 2028, with the potential to scale beyond 14x reticle in 2029. We believe TSMC understands the need to expand its advanced packaging capabilities to meet future HPC and AI requirements, as evidenced

by the company's multi-year roadmap progression from the previous 9.5x-reticle CoWoS. Nevertheless, scaling CoWoS beyond 10x reticle remains challenging. [Figure 3](#) illustrates how, as package size grows, fewer packages can be placed on a circular wafer, as compared to a square panel. This ultimately translates to lower yield with less known-good-dies (KGDs) per wafer. We believe CoPoS is a better solution for larger size package than CoWoS, particularly as TSMC could enhance the scalability of CoPoS through migrating to larger panel sizes such as 510x515mm².

5) Cost

Cost structure may also be an advantage of EMIB-T, given its more streamlined packaging process flow and removal of expensive interposer. That said, the ultimate cost competitiveness of the technology will depend on whether Intel and the substrate makers can achieve high-volume production with commercially viable yield rates.

In our view, yield and operational execution will be critical factors when customers evaluate the two packaging options. Lower yields can raise packaging costs, as well as impact the time to market and the pace of ramp. TSMC has a clear advantage in execution and manufacturing track record. The company has repeatedly demonstrated its ability to ramp new technologies to high-volume production, supported by its extensive experience serving a broad customer base across both front-end and advanced-packaging services.

If Intel's EMIB-T ramps, what's the impact on TSMC?

Based on our competitive analysis in the prior section, we view TSMC's offering as the stronger solution for package performance and integration. CoPoS could achieve denser, shorter interconnects, and a higher level of integration, including 3D stacking, more embedded silicon capacitors, and integration of co-packaged optics (CPO), in our view. In comparison, Intel's EMIB-T may offer advantages in packaging simplicity and process flow, and competitive cost (assuming reasonable yields). Intel's EMIB-T technology may also be easier to scale to very large package sizes (i.e. 14x reticle or larger), although we believe TSMC's roadmap of 510x515mm² CoPoS panel could narrow this advantage over time.

Overall, we believe TSMC remains the preferred platform for customers seeking the highest-performance packaging solution with the strongest execution track record. But for clients primarily focused on scaling package size to integrate more logic/HBM chips, Intel's EMIB-T could represent an attractive alternative. Beyond the technology, Intel's EMIB-T is also gaining traction because of TSMC's tight capacity and geopolitical concerns, which have encouraged customers to diversify supply sources.

Assuming CoPoS enters mass production in 2028, initially serving Nvidia's Feynman and expanding to other clients in 2029-30, we expect TSMC to retain its dominant position in advanced packaging. TSMC's edge is its strong capability to provide end-to-end solution, combining front-end process with a comprehensive technology portfolio of 2.5D/3D packaging and CPO. Intel is targeting mass production by H227. In the initial stages, we expect Intel to be selective in client engagement, prioritizing internal products and Google TPU v9, which is being developed by MediaTek and set for mass production in 2028, to ensure a smooth ramp and yield improvement.

To put things into perspective, in a stress test scenario, our analysis shows that TSMC could sustain a 50% revenue CAGR for advanced packaging in 2026-30E even if Intel gains meaningful share through EMIB-T within Google TPUs and captures modest volumes for other XPU's. This scenario assumes Intel ramps EMIB-T sales to US\$16bn, or 13% of industry share by 2030, with close to 24mn AI accelerate dies or ~6mn chipsets with four XPU's per package. Under this scenario, TSMC could sustain 65-70% share by 2030E vs 80-85% in 2026-27E. TSMC's share could further increase if Google's new TPUs (eg. v10) return to CoPoS and adopt CPO integration.

In our lower case scenario ([Figure 17](#)), we assume Intel's EMIB-T ramps more successfully, with Intel reaching an 18% share in the advanced packaging market by 2030E. Under this assumption, we estimate TSMC's share in advanced packaging could decline to 62% in 2030E, with 47% sales CAGR for advanced packaging 2026-30E.

Figure 16: Stress test scenario—TSMC's advanced packaging could sustain 65-70% share by 2030E, even if Intel ramps EMIB-T to support 24mn XPU dies or 6mn chipsets

(US\$mn)	2026E	2027E	2028E	2029E	2030E	26-30E CAGR
Advanced packaging industry revenue	19,025	36,300	58,080	87,120	121,968	59%
YoY	87%	91%	60%	50%	40%	
TSMC's advanced packaging sales	16,034	29,730	41,676	59,207	81,378	50%
YoY	80%	85%	40%	42%	37%	
% of TSMC total sales	9%	12%	13%	16%	18%	
Intel's EMIB-T sales for external			5,438	9,570	15,791	
Google TPU			5,003	8,374	13,159	
Units (mn)			2.3	3.5	5.0	
EMIB-T content (US\$)			2,175	2,393	2,632	
Others			435	1,196	2,632	
Units (mn)			0.2	0.5	1.0	
EMIB-T content (US\$)			2,175	2,393	2,632	
Advanced packaging industry share						
TSMC	84%	82%	72%	68%	67%	
Intel			9%	11%	13%	
OSATs	16%	18%	19%	21%	20%	

Source: UBS estimates

Figure 17: Lower case scenario - TSMC's share in advanced packaging could decline to 62% by 2030E, assuming better success of Intel's EMIB-T

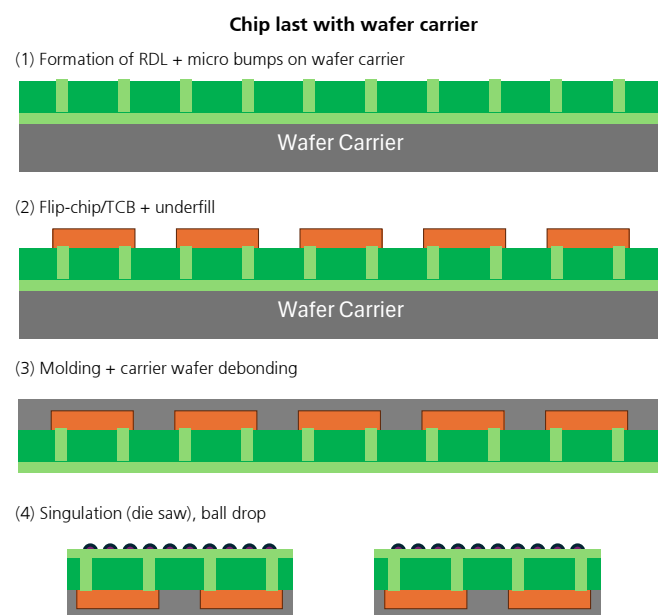
(US\$mn)	2026E	2027E	2028E	2029E	2030E	26-30E CAGR
Advanced packaging industry revenue	19,025	36,300	58,080	86,883	120,081	59%
YoY	87%	91%	60%	50%	38%	
TSMC's advanced packaging sales	16,034	29,730	41,676	56,578	74,227	47%
YoY	80%	85%	40%	36%	31%	
% of TSMC total sales	9%	12%	13%	15%	16%	
Intel's EMIB-T sales for external			5,438	11,963	21,054	
Google TPU			5,003	10,766	17,106	
Units (mn)			2.3	4.5	6.5	
EMIB-T content (US\$)			2,175	2,393	2,632	
Others			435	1,196	3,948	
Units (mn)			0.2	0.5	1.5	
EMIB-T content (US\$)			2,175	2,393	2,632	
Advanced packaging industry share						
TSMC	84%	82%	72%	65%	62%	
Intel			9%	14%	18%	
OSATs	16%	18%	19%	21%	21%	

Source: UBS estimates

Panel carriers and glass core substrates

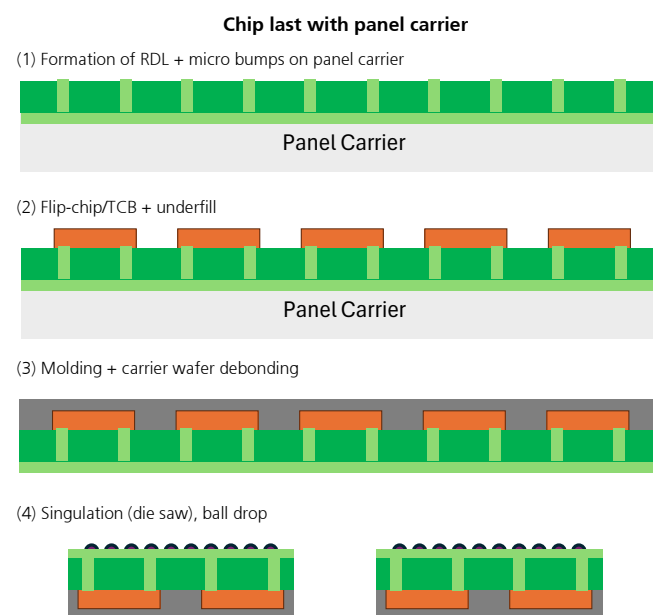
Given some market confusion on the role of glass in TSMC's CoPoS roadmap and the associated production risk, we believe it is important to clarify our view on both panel carriers and glass core substrates. Glass is relevant to CoPoS in two areas: 1) panel carriers and 2) glass core substrates. Intel's EMIB-T could also adopt glass core substrates over time. We believe that glass adoption (whether for panel carrier or glass-core substrate) should not be a major yield or production risk for the ramp of either TSMC's CoPoS or Intel's EMIB-T in 2028. Industry feedback indicates that both packaging technologies should be in mass production in 2028 without requiring glass-core substrates. In addition, if TSMC adopts glass panel carriers, we would not expect this to constrain CoPoS production or weigh on yields.

Figure 18: Wafer carrier chip last method



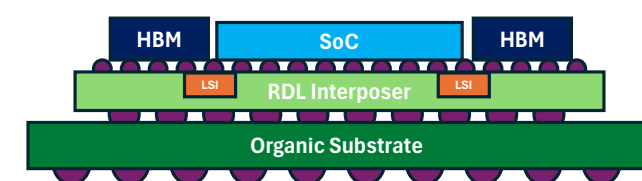
Source: UBS

Figure 19: Panel carrier chip last method



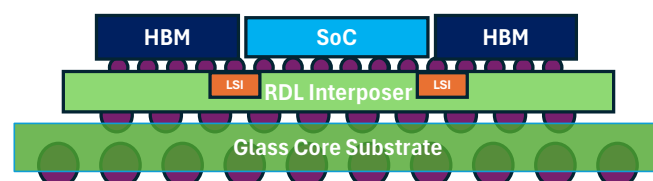
Source: UBS

Figure 20: Current package technology with organic substrate



Source: UBS

Figure 21: Future package technology with glass core substrate

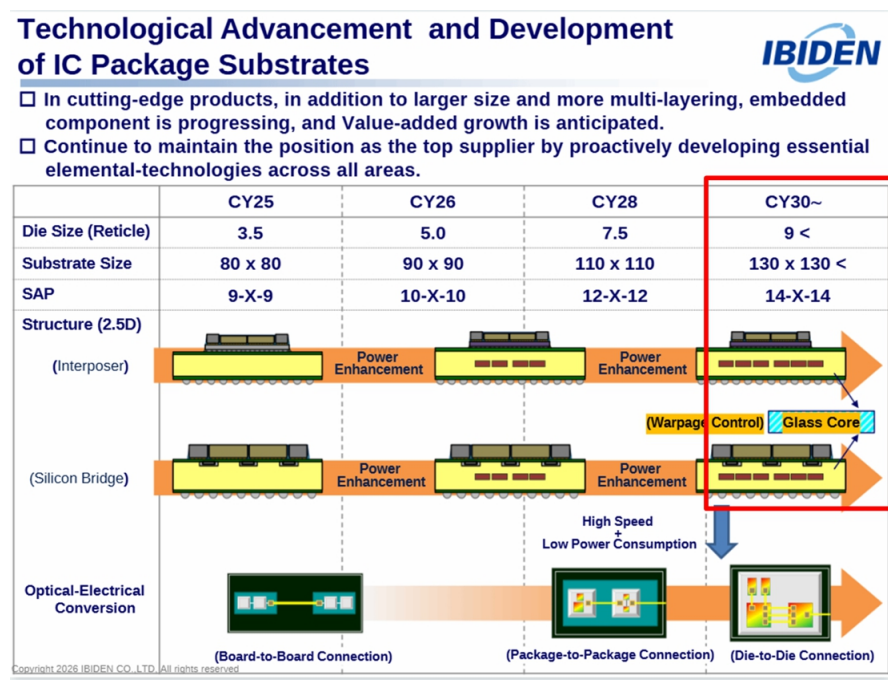


Source: UBS

What is a panel carrier? During the CoP packaging process, the logic and HBM chips are not attached until after the redistribution layer (RDL) interposer has been formed on the carrier panel, similar to CoWoS-L's chip-last process. The carrier panel serves as a temporary platform to support interposer fabrication and subsequent die attach processes. For CoPoS, we believe TSMC is evaluating different materials for carrier, but it appears glass is the leading candidate based on current visibility.

What is a glass-core substrate? Glass-core substrate could be viewed as a next generation ABF substrate. With glass materials at the core of the substrate, and wrapped with ABF layers to increase the rigidity of the substrate for larger package. Given the ongoing trend of expanding package sizes, the industry is investing heavily in glass-core substrate development. However, we believe the technology is still in the early stages and is unlikely to reach mass production before 2029-30E, if not later.

Figure 22: Ibiden glass-core substrate timeline at 2030+



Source: Company data (Ibiden May 2026 earnings presentation)

Challenges for glass-core substrates. Glass-core substrates offer significantly greater rigidity than conventional organic ABF substrates, which is crucial for package sizes larger than 14x reticle. Instead of the traditional TSV (through-silicon-vias), glass-core substrate fabrication adopts high-aspect-ratio TGV (through-glass-vias). This requires the development of advanced process equipment (e.g. for laser and etching) to enable accurate and precise drilling of the glass core materials. Compared with organic ABF substrates, glass is more rigid and brittle, making via formation substantially more difficult. Other production challenges include copper implanting for TGV holes, micro-cracks (if using glass as the core), and reliability issues. The supply chain will need to progress through a learning curve to improve yields, optimise process flows, and minimize cracks during the mass production stage.

Major supply chain beneficiaries

Figure 23: Notable CoPoS suppliers

TSMC CoPoS supply chain	Suppliers
Deposition and re-distribution layer	Applied Materials, SCREEN, Tokyo Electron, Canon
Baking and thermal treatment	APT, C Sun, GMM, Kokusai Electric
Wet process	GPTC, Scientech
Debonding	SÜSS MicroTec, Tokyo Electron, EVG, Scientech, E&R engineering, Daxin
CMP	Applied Materials, Disco, Ebara, LAM research, Tokyo Electron, GPM
Molding equipment	Apic Yamada, Towa
Pod	Gudeng, Chung King
Underfill dispenser	All Ring, Asymtek
Die attach	Kulicke & Soffa, ASMPT, Shibaaura
Metrology, inspection, and automation	KLA, Chroma, Camtek, Favite, Ta Liang Technology, V5 Technologies
OSAT partner	ASE/SPIL
Substrate	Ibiden, Unimicron, Toppan, Shinko, Nanya PCB, Kinsus

Source: Company data, UBS

Figure 24: Notable EMIB-T suppliers

Intel EMIB-T supply chain	Suppliers
Substrate	Ibiden, Unimicron
OSAT partner	Amkor
Die attach	Kulicke & Soffa, ASMPT

Source: Company data, UBS

Equipment

For TSMC's CoPoS, we believe most existing CoWoS equipment suppliers, along with several new entrants, will seek to capitalize on the next advanced-packaging capex cycle. TSMC began installing its pilot line in late Q2'26, and we expect the company to finalize the first-generation process flow, equipment specifications and key suppliers by mid-2027.

Major equipment providers include:

Die bonders (Sunny Lin): We believe TCB (thermal compression bonding) will be the preferred tool for larger reticle sized packages, given its yield advantages over mass reflow solutions. We anticipate CoPoS to use TCB for both the chip-on-panel (CoP) and on-substrate (oS) stages to better manage warpage challenges associated with larger packages. Major TCB suppliers to TSMC include ASMPT and K&S.

Wet process equipment (Sunny Lin): Wet process tools are used throughout the advanced packaging process to clean the surface of the die and interposer surfaces using chemical treatments. For CoPoS, we expect TSMC to largely rely on its existing CoWoS suppliers - GPTC and Scientech. Given its stronger technology capability, we believe GPTC could increase its market share to 70-80%, vs its roughly 50% share in CoWoS over recent years.

Underfill equipment (Sunny Lin): Underfill tools dispense and inject protective materials beneath post-bonded CoWoS chips and interposers, helping strengthen the package, improve reliability, and prevent damage from stress and temperature changes. Underfill is required for both the CoP and on-substrate stages. All Ring is currently TSMC's sole supplier of underfill equipment for on-substrate, and we anticipate it to remain the sole supplier for on-substrate in CoPoS, and to likely penetrate the CoP process as well.

Metrology (Jerry Su): Metrology tools are critical to measuring key dimensions and process parameters during the semiconductor manufacturing process to ensure production quality and maximize yields. Chroma has developed its proprietary BLIS technology, a high-speed contactless measurement system based on white-light interferometry that delivers sub-nanometer resolution. The company entered TSMC's supply chain in 2024, and its tools have been adopted for advanced-packaging applications (eg, CoWoS), including precise measurement of TSV/VIA, RDL structures, probe marks, overlay accuracy, and surface roughness. Chroma has also been working with customers on metrology solutions for CoPoS (310mm x 310mm substrate size) and received an initial order in H1'26.

Advanced packaging pods (Sunny Lin): Chung King in Taiwan is currently the leading supplier of advanced-packaging pods for TSMC's CoWoS production. As TSMC transitions toward CoPoS, we expect additional competitors to target this opportunity. Gudeng, leveraging its experience in mask pods and wafer FOUPs, appears to be among the most credible challengers seeking to penetrate the market, in our view.

Intel EMIB-T—A Simpler Equipment Set: As for Intel's EMIB-T, we believe the required equipment set may be simpler as it only involves chip-on-substrate assembly and does not require a separate chip-on-panel stage. As a result, a key leading indicator of EMIB-T capacity expansion is likely to be **thermo-compression bonding (TCB)** tool demand for die bonders. Intel has long favored TCB-based assembly, and its primary suppliers today include K&S and ASMPT. We also note that Intel may rely more on US-based equipment suppliers whereas TSMC's advanced packaging supply chain is more localized in Asia.

ABF & glass-core substrates

TSMC's CoPoS (Shingo Hirata, Diana Chang): We believe Ibiden is well positioned to benefit from CoPoS adoption given its leadership in high-end ABF substrates, with its current 70-80% market share in ABF substrate in NVIDIA Blackwell / Blackwell Ultra GPU. Ibiden has a strong track record as a substrate supplier for TSMC's advanced packaging, including CoWoS-L, and has demonstrated an ability to minimize substrate warpage, an important factor in maximizing yield during TSMC's on substrate (oS) process. Given the anticipated shortage of high-end ABF substrates over the coming years, customers may increasingly pursue multi-sourcing strategies. However, we expect

Ibiden to maintain a strong competitive position as reliability and manufacturing track record are likely to outweigh cost considerations, given ABF package costs represent only a small portion of the overall bill of materials for GPU/HBM.

Among Taiwan substrate makers, we believe Unimicron is a key beneficiary of CoPoS deployment given its investments in high-end ABF substrate capacity and technology. Nan Ya PCB (NYPECB) has also recently announced capacity expansion with a focus on CoPoS.

Intel's EMIB-T (Shingo Hirata, Diana Chang): We believe Ibiden should also benefit from Intel's EMIB-T as the top supplier. EMIB-T requires substrate makers to integrate silicon bridge into ABF substrate. EMIB-T's key upgrade is that it builds TSV in the silicon bridge for vertical power delivery. To enable vertical power delivery, the process needs to build metal contacts at both the top and bottom of the bridge cavities and precisely position the silicon bridge to establish the power delivery route. These process steps do not exist in conventional ABF substrate manufacturing and introduces significant additional complexity. At the same time, EMIB-T streamlines the overall packaging process. Given the higher complexity and additional benefits for customers, substrate makers can enjoy much higher value add relative to standard ABF substrate. Given Ibiden's strong track record in high end package and its experience with previous generations of EMIB, we expect the company to maintain a dominant position in the EMIB-T supply chain.

Ibiden announced a ¥220bn investment for its Gama plant in February 2026, with mass production for EMIB-T scheduled to begin in late 2027 and a more meaningful ramp expected in 2028. Our checks suggest that demand from Google TPU programs alone could largely absorb the planned capacity, increasing the likelihood of additional capex announcements as demand visibility improves.

Unimicron previously partnered with Intel on prior generations of EMIB, though at relatively low volumes. The company is also developing EMIB-T capabilities, although yields remain below optimal levels given the manufacturing challenges. We believe Ibiden will be the primary supplier for Intel's EMIB-T, with Unimicron potentially serving as a second source if volumes exceed expectations.

OSAT partners

TSMC's CoPoS (Sunny Lin, Randy Abrams): We expect TSMC to maintain a collaborative manufacturing model with OSAT partners once CoPoS reaches high-volume production. Under this framework, TSMC would likely focus on expanding chip-on-panel (CoP) capacity, while OSATs support the chip-on-substrate (oS) assembly stage. For Taiwan-based expansion, we expect ASE to be TSMC's primary OSAT supplier. In Arizona, Amkor appears to be the most likely on-substrate partner.

Intel's EMIB-T (Randy Abrams): We view Intel vertically integrating its EMIB-T packaging for external customers in its Penang Malaysia plant currently under-going an expansion. Intel and Amkor also signed a strategic partnership in 2025 making available Amkor's Korea, Portugal and US lines for EMIB assembly. We believe Intel may collaborate to outsource some of its own CPU products for on substrate packaging to Amkor, in our view.

Intel should prioritize focus on ramping up the key external products like Google TPUs in 2028-29 at its own facilities with vertical integration to ensure delivery while outsourcing the more mature in-house products where it can directly work with its outsource partner to qualify the process. More broadly, we see potential for outsourcing of certain steps like C4 wafer bumping for power, grounding and low speed signals between the die and package wafer bump where the industry is in tight supply.

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2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
ASE Industrial ^{16,28}	3711.TW	Buy	NT\$613.00	24 Jul 2026
ASMPT Limited ^{13,28}	0522.HK	Buy	HK\$163.40	24 Jul 2026
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